zetesis-puremath Blueprint

A Lean 4 pure-mathematics reservoir:
analytic measurability, information theory, concentration, minimax,
kernel embeddings, and dual VC.

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Contents

Chapter 1

Introduction

This blueprint pins every mathematical definition and headline theorem in `zetesis-puremath`[1] to its Lean 4 declaration. The companion source is a personal pure-math reservoir: content needed by the author's formalization projects (`formal-learning-theory-kernel`, SMFE, and others) that is either absent from Mathlib or specialized in a way Mathlib's current formulation does not support directly.

1.1 What is formalized here

Six substrates, each with its own chapter below:

- **Measure theory on Polish spaces.** Analytic sets are null-measurable under finite Borel measures; Kechris 30.13 capacitability theorem for analytic sets; real-valued metric total variation distance.
- **Information theory.** Real-valued Kullback-Leibler bridge to Mathlib's `ENNReal`-valued `klDiv`; binary KL with a full derivative / monotonicity chain; Pinsker's inequality with the sharp constant; mutual information and its zero-iff-product characterization.
- **Probability.** `BoundedRandomVariable` typeclass with a Chebyshev-via-majority bound; double-sample / `ValidSplit` / `SplitMeasure` exchangeability infrastructure; `FintypePMF` (an $\mathbb{R}$-valued, `linarith`- compatible specialization of Mathlib's `PMF`).
- **Decision theory.** Covering-based minimax; multiplicative-weights-update (MWU) in the Freund-Schapire $\beta = (1 - \eta)$ convention; approximate-minimax bound $T = \mathcal{O}(\log N/\varepsilon^2)$.
- **Reproducing-kernel Hilbert spaces.** Kernel mean embedding with the reproducing property; Maximum Mean Discrepancy (MMD) and characteristic kernels; Hilbert-Schmidt Independence Criterion (HSIC).
- **Combinatorics.** Assouad's 1983 dual VC bound $\mathrm{VCdim}(M^\top) \le 2^{d+1} - 1$.

1.2 Verification tiers

All 114 public theorems discharge under the following tiers, all green at commit `471e5d5` (see `test/verification/CI/`):

- **Tier 0**: `lake build` — 0 errors, 0 warnings, 0 `sorry` in `ZPM/`.

[1] `https://github.com/Zetetic-Dhruv/zetesis-puremath`

- **Tier 1**: `#print axioms` — 113 theorems use only `propext`, `Classical.choice`, `Quot.sound`; the 114th (`challenge_bnd_subset_cyl`) uses none.
- **Tier 2**: `leanchecker -fresh` per module (GitHub Actions `workflow_dispatch`).
- **Tier 3**: `comparator` + Landlock sandbox on GitLab CI; *"Lean default kernel accepts the solution. Your solution is okay!"* (pipeline #2463996501, duration 36 min).

Every statement in this blueprint carries a `\lean{...}` annotation linking to the Lean declaration; where the full proof is in the source, `\leanok` marks the entry as formally verified.

Chapter 2

Measure theory

2.1 Analytic measurability

Theorem 2.1 (Analytic null-measurability). *Lean:* `MeasureTheory.AnalyticSet.nullMeasurableSet`

Let α be a Polish space, μ a finite Borel measure on α, and $S \subseteq \alpha$ analytic. Then S is μ-null-measurable.

Theorem 2.2 (Inner approximation by compacts). *Lean:* `MeasureTheory.AnalyticSet.exists_isCompact_measureReal_`

Under the hypotheses of Theorem 2.1, for every $\delta > 0$ there exists a compact $K \subseteq S$ with $\mu(S) - \mu(K) < \delta$ (in $\mathbb{R}$-valued form).

2.2 Choquet capacity

Definition 2.3 (Choquet capacity). Lean: `MeasureTheory.IsChoquetCapacity`

A map $I : \mathrm{Set}(\alpha) \to \mathbb{R}^{\infty}_{\geq 0}$ is a *Choquet capacity* iff (i) it is monotone, (ii) continuous from below along increasing sequences, and (iii) continuous from above along antitone sequences of closed sets. Every finite Borel measure on a Polish space is a Choquet capacity [DM78].

Definition 2.4 (Compact capacity hull). Lean: `MeasureTheory.compactCap`

For μ and S as above,

$$\mathrm{compactCap}\, \mu\, S \;:=\; \sup\{\mu(K) : K \subseteq S,\ K \text{ compact}\}.$$

Theorem 2.5 (Kechris 30.13: analytic capacitability). *Lean:* `MeasureTheory.AnalyticSet.cap_eq_iSup_isCompact`

Let I be a Choquet capacity on a Polish space α, and S an analytic subset of α. Then

$$I(S) \;=\; \sup\{I(K) : K \subseteq S,\ K \text{ compact}\}.$$

The proof is the Kechris 30.13 diagonal construction on cylinders in Baire space $\mathbb{N}^{\mathbb{N}}$ [Kec95]. Together with Definition 2.4 and the fact that every finite Borel measure is a Choquet capacity, this specializes to $\mu(S) = \mathrm{compactCap}\, \mu\, S$ (Theorem 2.1 is an immediate corollary).

2.3 Baire-space cylinder infrastructure

The proof of Theorem 2.5 requires a small theory of cylinders in $\mathbb{N}^{\mathbb{N}}$:

Definition 2.6 (Cylinder, bounded box, truncate). Lean: `Cyl`, `Bnd`, `truncate`

For $N : \mathbb{N} \to \mathbb{N}$ and $n \in \mathbb{N}$, $\mathrm{Cyl}\, N\, n \;:=\; \{g : \mathbb{N} \to \mathbb{N} \mid \forall i \leq n,\ g\, i \leq N\, i\}$ and $\mathrm{Bnd}\, N \;:=\; \bigcap_n \mathrm{Cyl}\, N\, n$. $\mathrm{Bnd}\, N = \prod_i [0, N_i]$ is compact. The map $\mathrm{truncate}\, N\, g\, i := \min(g\, i, N\, i)$ sends any sequence into $\mathrm{Bnd}\, N$ and agrees with g on cylinders.

The lemmas `bnd_subset_cyl`, `cyl_succ_eq`, `cyl_inter_eq_cyl_update`, `monotone_cyl_split`, `iInter_closure_image_cyl_eq`, `isCompact_bnd`, and `truncate_mem_bnd` are the technical core of the proof; they are annotated with `\lean` in the source repository and are all `\leanok`.

2.4 Total variation distance

Definition 2.7 (Real-valued total variation). Lean: `MeasureTheory.tvDistReal`

For finite Borel measures P, Q on α,

$$\mathrm{TV}(P, Q) \ := \ \sup_A \lvert P(A).\mathrm{toReal} - Q(A).\mathrm{toReal}\rvert$$

where the supremum is over measurable $A \subseteq \alpha$. This is the *half-sup* / metric convention (Billingsley, Villani), chosen so that Pinsker reads as $\mathrm{TV}(P, Q) \le \sqrt{\mathrm{KL}(P\|Q)/2}$. The lemmas `tvDistReal_set_nonempty` and `tvDistReal_set_bddAbove` discharge the nonemptiness and boundedness conditions that $\mathbb{R}$-valued `sSup` requires.

Remark 2.8. The repository also contains a second total-variation object, FintypePMF.tvDistance $p\,q = \sum_a |p\,a - q\,a|$ (the L^1 convention, see Chapter 4); the two conventions differ by a factor of 2 on discrete supports. Downstream consumers should not compose them without inserting the factor-of-$\frac{1}{2}$ correction.

Chapter 3

Information theory

3.1 Real-valued KL bridge

Definition 3.1 (Real-valued KL). Lean: `InformationTheory.klDivReal`

For measures P, Q on a measurable space, set

$$\mathrm{KL}_{\mathbb{R}}(P, Q) \;:=\; \begin{cases} \int \log \frac{\mathrm{d}P}{\mathrm{d}Q}\,\mathrm{d}P & \text{if } P \ll Q, \\ 0 & \text{otherwise.} \end{cases}$$

This bridges Mathlib's `ENNReal`-valued `klDiv` to $\mathbb{R}$ for `linarith`-compatible arithmetic; the identity $\mathrm{KL}_{\mathbb{R}}(P, Q) = (\mathrm{KL}(P, Q)).\mathrm{toReal}$ holds unconditionally (`klDivReal_eq_toReal_klDiv`).

Theorem 3.2 (Non-negativity). *Lean:* `InformationTheory.klDivReal_nonneg`

$\mathrm{KL}_{\mathbb{R}}(P, Q) \geq 0$ *for all measures (Gibbs' inequality).*

Theorem 3.3 (Zero iff equal, under finiteness). *Lean:* `InformationTheory.klDivReal_eq_zero_iff`

If $P \ll Q$, $\mathrm{KL}(P, Q) \neq \infty$, *and both are finite measures, then* $\mathrm{KL}_{\mathbb{R}}(P, Q) = 0 \iff P = Q$ *[CT06, Thm 2.6.3].*

3.2 Binary KL and Pinsker's inequality

Definition 3.4 (Binary KL). Lean: `InformationTheory.klBin`

For $p, q \in [0, 1]$,

$$\mathrm{KL}_{\mathrm{bin}}(p, q) \;:=\; p \log \tfrac{p}{q} + (1-p) \log \tfrac{1-p}{1-q}$$

with the convention $0 \cdot \log(0/\cdot) = 0$.

Lemma 3.5 (Derivative in q). *Lean:* `InformationTheory.hasDerivAt_klBin_q`

For $q \in (0, 1)$, $\partial_q \mathrm{KL}_{\mathrm{bin}}(p, q) = (q - p)/(q(1-q))$.

Theorem 3.6 (Binary Pinsker, sharp constant 2). *Lean:* `InformationTheory.binary_pinsker`

For $p \in [0, 1]$ *and* $q \in (0, 1)$, $2(p-q)^2 \;\leq\; \mathrm{KL}_{\mathrm{bin}}(p, q)$.

Theorem 3.7 (Pinsker's inequality). *Lean:* `InformationTheory.pinsker_proof`

For probability measures $P \ll Q$ *with* $\mathrm{KL}(P, Q) < \infty$,

$$\mathrm{TV}(P, Q) \;\leq\; \sqrt{\mathrm{KL}_{\mathbb{R}}(P, Q)/2}$$

[CT06, Thm 11.6.1]. The convention matches Definition 2.7 (half-sup TV); with the L^1 *convention the bound reads* $\|P - Q\|_1 \leq \sqrt{2\,\mathrm{KL}}$.

Theorem 3.8 (Binary-to-general coarsening). *Lean:* `InformationTheory.klBin_le_klDivReal`

For every measurable A with $P(A) = p$, $Q(A) = q$, $\mathrm{KL}_{\mathrm{bin}}(p,q) \leq \mathrm{KL}_{\mathbb{R}}(P,Q)$. This is the two-point data-processing inequality and is the step linking the binary Pinsker to the general measure-theoretic Pinsker.

3.3 Mutual information

Definition 3.9 (Real-valued mutual information). Lean: `InformationTheory.mutualInformationReal`

For a probability measure P on $X \times Y$ with marginals P_X, P_Y, $\mathrm{MI}_{\mathbb{R}}(P) := \mathrm{KL}_{\mathbb{R}}(P, P_X \otimes P_Y)$.

Theorem 3.10 (Non-negativity and independence). *Lean:* `InformationTheory.mutualInformationReal_nonneg`, `InformationTheory.mutualInformationReal_eq_zero_iff_product`

$\mathrm{MI}_{\mathbb{R}}(P) \geq 0$*; under* $P \ll P_X \otimes P_Y$ *and finiteness,* $\mathrm{MI}_{\mathbb{R}}(P) = 0 \iff P = P_X \otimes P_Y$.

Chapter 4

Probability

4.1 Bounded random variables + concentration

Definition 4.1 (Bounded random variable). Lean: `ProbabilityTheory.BoundedRandomVariable`

$f : \Omega \to \mathbb{R}$ is *bounded* in $[a, b]$ under μ iff f is measurable and $f(\omega) \in [a, b]$ for μ-almost every ω.

Theorem 4.2 (Chebyshev majority bound). *Lean:* `ProbabilityTheory.chebyshev_majority_bound`

Let $X_1, \ldots, X_k$ be independent random variables with each $\mu(A_j) \geq 2/3$ where $A_j = \{X_j = \text{correct}\}$. Then for $k \geq 9/\delta$, the probability that a majority of the X_j is correct is at least $1 - \delta$. Proof via Popoviciu's variance bound + Chebyshev's inequality [BLM13, §2].

4.2 Exchangeability and double-sample

Definition 4.3 (Sample bundle). Lean: `ProbabilityTheory.ExchangeableSample`

A bundle $\langle m, \mu, \mathrm{IsProbabilityMeasure}\, \mu \rangle$ where $m \in \mathbb{N}$ is the sample size and μ is a probability measure on the sample space. Used with the product measure $\mu^{\otimes 2m}$ as the natural exchangeable double-sample domain.

Definition 4.4 (Valid split). Lean: `ProbabilityTheory.ValidSplit`

$s : \mathrm{Fin}(2m) \to \mathrm{Bool}$ is a *valid split* iff exactly m of its values are `true`. There are $\binom{2m}{m}$ valid splits (Vapnik-Chervonenkis double-sample [VC71]).

Definition 4.5 (Uniform split measure). Lean: `ProbabilityTheory.SplitMeasure`

The uniform probability measure on the set of valid splits. This is the symmetrization measure for the double-sample trick.

4.3 Finite-type PMF

Definition 4.6 (Real-valued fintype PMF). Lean: `ProbabilityTheory.FintypePMF`

For a fintype α, a `FintypePMF` α is a pair $\langle p, h_{\geq 0}, h_{\sum = 1} \rangle$ where $p : \alpha \to \mathbb{R}$, $p \geq 0$ pointwise, and $\sum_{a:\alpha} p(a) = 1$. This is an $\mathbb{R}$-valued, `linarith`-compatible specialization of Mathlib's $\mathbb{R}^{\infty}_{\geq 0}$-valued `PMF` [BLM13, §4].

Theorem 4.7 (Bridge to Mathlib PMF). *Lean:* `ProbabilityTheory.FintypePMF.toPMF_apply`

The map $\mathrm{toPMF} : \mathrm{FintypePMF}\, \alpha \to \mathrm{PMF}\, \alpha$ *satisfies* $(\mathrm{toPMF}\, p)(a) = \mathrm{ENNReal.ofReal}(p.\mathrm{prob}\, a)$, *which is lossless on valid FintypePMF inputs.*

Definition 4.8 (L^1 TV distance). Lean: `ProbabilityTheory.FintypePMF.tvDistance`

tvDistance $p\,q \;:=\; \sum_{a:\alpha} |p(a) - q(a)|$. This is the *full* L^1 convention, not the half-sup TV of Definition 2.7; on discrete supports they differ by a factor of 2.

Theorem 4.9 (Basic properties). *Lean: `ProbabilityTheory.FintypePMF.tvDistance_nonneg`, `ProbabilityTheory.Fintyp`*
`ProbabilityTheory.FintypePMF.tvDistance_self`

tvDistance *is non-negative, symmetric, and* 0 *on* $p = q$.

Theorem 4.10 (Transfer principle). *Lean: `ProbabilityTheory.FintypePMF.expectation_approx_of_tv`*

For Boolean-valued f, $|\mathbb{E}_p[f] - \mathbb{E}_q[f]| \;\leq\;$ tvDistance(p, q). *In the half-sup TV convention this would read* $\leq 2 \cdot \mathrm{TV}(p, q)$.

Chapter 5

Decision theory: minimax and MWU

5.1 Boolean-matrix zero-sum games

Definition 5.1 (Boolean-matrix payoff). Lean: `ProbabilityTheory.boolGamePayoff`

For a Boolean matrix $M : R \to C \to \mathrm{Bool}$, a row distribution $p : \mathrm{FintypePMF}\, R$, and a pure column $c : C$,

$$\mathrm{boolGamePayoff}\, M\, p\, c \ := \ \sum_{r:R} p(r) \cdot [M\, r\, c].$$

This is the expected $\{0,1\}$-gain of the row player, against a pure column play, in the matrix game M [FS99].

Theorem 5.2 (One-sided minimax sanity). *Lean: `ProbabilityTheory.minimax_value_le_one`*

If the row player can guarantee value v against every mixed column (the `hrow` hypothesis), then $v \leq 1$.

5.2 Multiplicative weights update

Definition 5.3 (MWU config). Lean: `ProbabilityTheory.MWUConfig`

An `MWUConfig` C is a pair $\langle w, h_{>0} \rangle$ where $w : C \to \mathbb{R}$ is the strictly positive weight vector. The *potential* is $\Phi(w) := \sum_c w(c)$, the *induced PMF* is $\mathrm{toPMF}(w)(c) := w(c)/\Phi(w)$ [AHK12, Thm 2.1].

Definition 5.4 (Update and closed-form). Lean: `ProbabilityTheory.mwuUpdateWeights`

Given a row $r : R$ and step size $\eta < 1$, the update is $w'(c) := w(c) \cdot (1 - \eta)$ if $M\, r\, c = \mathrm{true}$, else $w'(c) := w(c)$. This is the Freund-Schapire $\beta = (1 - \eta)$ Hedge convention [FS97]. The closed form

$$w_T(c) \ = \ w_0(c) \cdot (1 - \eta)^{\mathrm{hitCount}_T(c)}$$

is Theorem `mwu_weight_eq_pow_hitCount`.

Theorem 5.5 (One-step potential bound). *Lean: `ProbabilityTheory.potential_one_step_bound`*

If the row player can guarantee value v against every mixed column and $0 \leq \eta < 1$, then after one MWU step $\Phi_{t+1} \leq \Phi_t \cdot (1 - \eta v)$.

Theorem 5.6 (T-step potential bound). *Lean: `ProbabilityTheory.mwu_potential_T_bound`*

Induction gives $\Phi_T \ \leq \ |C| \cdot (1 - \eta v)^T$.

5.3 Approximate minimax

Theorem 5.7 (Approximate minimax via MWU). *Lean:* `ProbabilityTheory.mwu_approx_minimax`

Fix $\varepsilon > 0$, set $\eta = \varepsilon/4$ and $T = \max\{1, \lceil 16 \log N/\varepsilon^2 \rceil\}$ where $N = |C|$. If the row player can guarantee value v against every mixed column, then the empirical row mixture produced by T rounds of MWU achieves

$$v - \varepsilon \;\leq\; \text{boolGamePayoff}\, M\, p\, c \qquad \textit{for every pure } c \in C.$$

This is the Freund-Schapire constructive minimax [FS99, Thm 3]; [AHK12, §3.1] unifies this with general online learning.

Theorem 5.8 (Covering-based minimax (existential)). *Lean:* `ProbabilityTheory.covering_minimax`, `ProbabilityTheory.finite_approx_minimax`

An existence counterpart to Theorem 5.7: given the same row-guarantees hypothesis, there is a mixed-row PMF with payoff at least $1/|C|$ against every pure column. This gives $v - \varepsilon \leq 1/|C|$ under a feasibility side-condition and is the logical dual of the quantitative MWU bound.

Chapter 6

Reproducing-kernel Hilbert spaces

Background conventions: the Mathlib class `RKHS H X` is the scalar-valued reproducing-kernel Hilbert space on a domain X, with Hilbert space $\mathcal{H} = H$ and feature map $k_x := \mathrm{kerFun}\, H\, x\, 1 \in H$ satisfying $f(x) = \langle k_x, f \rangle$ [Aro50]. Throughout, $\mathcal{H}$ denotes a complete, separable, real inner product space.

6.1 Bounded kernels

Definition 6.1 (Bounded kernel). Lean: `BoundedKernel`

A `BoundedKernel` $\mathcal{H}$ is a witness $\langle C, h_{\geq 0}, h_{\leq} \rangle$ with $C \geq 0$ and $\|k_x\| \leq C$ for all $x : X$. Bounded-feature kernels are the canonical setting for kernel mean embeddings [SC08, §4.2].

Theorem 6.2 (Evaluation bound). *Lean: `norm_apply_le_bound_mul_norm`*

$|f(x)| \leq C \cdot \|f\|$ for every $f \in \mathcal{H}$ and $x \in X$.

6.2 Kernel mean embedding

Definition 6.3 (Kernel mean embedding). Lean: `kernelMeanEmbedding`

For a measure P on X, set

$$\mu_P \ := \ \int_X k_x \, \mathrm{d}P(x) \ \in \ \mathcal{H}$$

(Bochner integral) [SGSS07].

Theorem 6.4 (Reproducing property for the embedding). *Lean: `kernelMeanEmbedding_reproducing`*

For integrable f, $\langle \mu_P, f \rangle \ = \ \int_X f(x) \, \mathrm{d}P(x)$ [GBR+12, §2, eq. (2)].

Theorem 6.5 (Norm bound). *Lean: `kernelMeanEmbedding_norm_le`*

Under `IsProbabilityMeasure` P, $\|\mu_P\| \leq C$.

6.3 Maximum Mean Discrepancy

Definition 6.6 (Squared MMD). Lean: `mmdSq`

$\mathrm{MMD}^2(P, Q) \ := \ \|\mu_P - \mu_Q\|^2$ [GBR+12, Defn. 2].

Theorem 6.7 (Metric axioms mod characteristic). *Lean: `mmdSq_nonneg`, `mmdSq_self`, `mmdSq_comm`*

$\mathrm{MMD}^2 \geq 0$, $\mathrm{MMD}^2(P, P) = 0$, and MMD^2 is symmetric in its arguments.

Definition 6.8 (Characteristic RKHS)**.** Lean: `IsCharacteristic`

$\mathcal{H}$ is *characteristic on* X iff the kernel mean embedding $P \mapsto \mu_P$ is injective. This is the modern definition [SGF+10, Defn. 6].

Theorem 6.9 (Zero-iff-equal under characteristic kernel)**.** *Lean:* `mmdSq_zero_iff`

Under `IsCharacteristic` $\mathcal{H}$*,* $\mathrm{MMD}^2(P, Q) = 0 \iff P = Q$*.*

6.4 Hilbert-Schmidt Independence Criterion

Definition 6.10 (HSIC via product-space MMD)**.** Lean: `hsicDef`

For a joint measure P on $X \times Y$ with marginals P_X, P_Y,

$$\mathrm{HSIC}(P) \ := \ \mathrm{MMD}^2(P,\ P_X \otimes P_Y).$$

This is the modern unified formulation [SGF+10, Thm 24]; originally introduced as a cross-covariance-operator Hilbert-Schmidt norm [GBSS05].

Theorem 6.11 (Zero-iff-independent)**.** *Lean:* `hsicDef_nonneg`, `hsicDef_eq_zero_of_independent`, `hsicDef_zero_iff_i`

$\mathrm{HSIC}(P) \geq 0$*;* $P = P_X \otimes P_Y$ *implies* $\mathrm{HSIC}(P) = 0$*; conversely, under characteristic kernels on* $X \times Y$*,* $\mathrm{HSIC}(P) = 0 \iff P = P_X \otimes P_Y$*.*

Chapter 7

Combinatorics: dual VC

7.1 Binary matrix setup

Definition 7.1 (Binary matrix). Lean: `BinaryMatrix`

$\mathrm{BinaryMatrix}\, m\, n := \mathrm{Fin}\, m \to \mathrm{Fin}\, n \to \mathrm{Bool}$. Type-definitionally equal to $\mathrm{Matrix}\,(\mathrm{Fin}\, m)\,(\mathrm{Fin}\, n)\,\mathrm{Bool}$.

Definition 7.2 (Shattering). Lean: `BinaryMatrix.shatters`, `BinaryMatrix.toFinsetFamily`

For a binary matrix M and $S \subseteq \mathrm{Fin}\, n$, M *shatters* S iff every pattern $t \subseteq S$ is realized by some row. The induced set family $\mathrm{toFinsetFamily}\, M$ consists of the true-supports of the rows of M [VC71, Sau72].

Theorem 7.3 (Consistency with `Finset.Shatters`). *Lean:* `BinaryMatrix.shatters_iff`

M shatters S $\iff$ $\mathrm{toFinsetFamily}\, M$ shatters S in Mathlib's sense.

7.2 Sauer-Shelah corollary

Theorem 7.4 (Cardinality of the induced family). *Lean:* `BinaryMatrix.card_toFinsetFamily_le`

If $\mathrm{VCdim}(\mathrm{toFinsetFamily}\, M) \le d$ then $|\mathrm{toFinsetFamily}\, M| \le \sum_{k=0}^{d} \binom{n}{k}$. This is the Sauer-Shelah bound [Sau72, VC71] routed through Mathlib's `Finset.card_shatterer_le_sum_vcDim` *(Pajor + Sauer-Shelah) into the matrix setting.*

7.3 Assouad's dual VC bound

Lemma 7.5 (Dual-to-primal shattering coding). *Lean:* `BinaryMatrix.transpose_shatters_imp_shatters`

If $M^\top$ shatters $S \subseteq \mathrm{Fin}\, m$ with $|S| \ge 2^{d+1}$, then M shatters some $T \subseteq \mathrm{Fin}\, n$ with $|T| = d + 1$. Proof by a bitstring-coding argument: over $(d+1)$-bit patterns on S, each pattern must be realized by a distinct column, giving $d + 1$ distinguishing columns that M shatters [Ass83, Thm 2.13].

Theorem 7.6 (Assouad's bound: $\mathrm{VCdim}(M^\top) \le 2^{d+1} - 1$). *Lean:* `BinaryMatrix.assouad_transpose_vcDim`

If $\mathrm{VCdim}(M) \le d$ then $\mathrm{VCdim}(M^\top) \le 2^{d+1} - 1$ [Ass83, Thm 2.13]. The bound is tight at small d; no strengthening is known for general d without additional structure.

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